Capstone Project Proposal: Predicting Starbucks Rewards Offer Response

Domain Background

Customer relationship management and personalized marketing are central problems in retail analytics. Loyalty programs such as Starbucks Rewards allow companies to observe customer interactions over time and deliver targeted promotions through mobile, email, web, and social channels. The business value of these programs depends on sending the right offer to the right customer at the right time.

Promotion targeting is a machine learning problem because customers differ in demographics, purchasing habits, tenure, and responsiveness to communication channels. A generic campaign may create unnecessary discount costs by sending offers to customers who would have purchased anyway, or it may miss customers who would have responded to a better matched offer. Predictive modeling can help estimate the likelihood that a customer will respond to a specific offer before the offer is sent.

This project uses the simulated Starbucks Rewards mobile app dataset provided in the Udacity Machine Learning Engineer Nanodegree. The dataset contains customer demographic records, offer metadata, and a time-ordered transcript of transactions and offer events. Although simulated, it reflects a realistic retail personalization problem: using historical app behavior and customer attributes to improve future offer targeting.

References:

- Udacity. *Starbucks Capstone Challenge Dataset*. Machine Learning Engineer Nanodegree.
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45, 5-32.
- Kohavi, R., & Provost, F. (1998). Glossary of terms. *Machine Learning*, 30, 271-274.

Problem Statement

The problem is to predict whether a customer will successfully respond to a Starbucks offer after receiving it.

Each record in the modeling dataset will represent one `offer received` event. The target variable will be binary:

- `1`: the customer successfully responded to the offer within its validity window.
- `0`: the customer did not successfully respond within the offer validity window.

A successful response is defined according to offer type:

- For BOGO and discount offers, the customer must view the offer and complete that same offer before it expires.
- For informational offers, the customer must view the offer and then make a transaction before it expires.

The project is quantifiable and replicable because the input data, response-labeling rules, feature engineering steps, train/test split, models, and evaluation metrics can all be implemented deterministically in a notebook.

Datasets and Inputs

The project uses three JSON files.

Dataset	Rows	Description
`portfolio.json`	10	Offer metadata, including offer id, type, reward, difficulty, duration, and delivery channels.
`profile.json`	17,000	Customer demographic data, including customer id, age, gender, income, and membership date.
`transcript.json`	306,534	Event-level customer activity, including transactions, offers received, offers viewed, and offers completed.

The relevant fields include:

- Customer features: age, gender, income, membership date.
- Offer features: offer type, difficulty, reward, duration, and channels.
- Event features: person id, event type, event time, transaction amount, and offer id.

The profile dataset contains missing demographic records represented by `age = 118`, missing `gender`, and missing `income`. These records will be removed from the primary demographic model because the project goal depends on demographic targeting.

The transcript data will be transformed into one row per offer exposure. For each received offer, future events for the same customer will be searched within the offer validity window to determine whether the offer was viewed and whether the customer completed or transacted after viewing.

Solution Statement

The proposed solution is a supervised binary classification model that predicts the probability of successful offer response for a customer-offer pair.

The solution workflow will:

1. Load and inspect the three JSON datasets.
2. Clean demographic records and normalize nested transcript fields.
3. Engineer customer, offer, membership, and channel features.
4. Construct a binary response label for every received offer.
5. Split the data into training and test sets using stratification.
6. Train benchmark and machine learning models.
7. Tune the strongest model using cross-validation.
8. Evaluate final performance on a held-out test set.
9. Analyze which demographic segments respond best to each offer type.

The final model can be used to score potential customer-offer combinations. In a business workflow, Starbucks could select the offer with the highest predicted response probability for each customer or suppress offers for customers with low predicted response probability.

Benchmark Model

The benchmark model will be a majority-class `DummyClassifier`. This model always predicts the most frequent class in the training data.

This benchmark is appropriate because it establishes the minimum useful performance threshold. If the response class is imbalanced, a simple model may achieve moderate accuracy while failing to identify successful responders. Therefore, the proposed machine learning model must improve substantially over the dummy classifier, especially on F1 score.

The benchmark will be evaluated on the same held-out test set as the final model.

Evaluation Metrics

The primary evaluation metric will be F1 score.

F1 score is appropriate because the business problem requires balancing precision and recall:

- Precision measures how often predicted responders actually respond.
- Recall measures how many actual responders the model identifies.
- F1 combines both into a single metric.

Accuracy will also be reported for interpretability, but it will not be the primary metric because class imbalance can make accuracy misleading.

ROC-AUC will be reported to evaluate how well the model ranks likely responders above unlikely responders across classification thresholds.

The evaluation metrics are:

Metric	Purpose
F1 score	Primary measure for balancing responder precision and recall.
Accuracy	General classification correctness.
ROC-AUC	Ranking quality across thresholds.
Confusion matrix	Error analysis by predicted and actual class.

Project Design

The project will be implemented in a Jupyter notebook using Python, pandas, NumPy, scikit-learn, Matplotlib, and Seaborn.

Data Preparation

The raw datasets will be loaded from JSON files. The portfolio data will be expanded so each delivery channel becomes a binary feature. The profile data will be cleaned by removing invalid demographic records and converting membership date into tenure-based features. The transcript data will be normalized by extracting offer ids and transaction amounts from nested dictionaries.

Label Construction

The transcript will be converted into a supervised learning table. For each `offer received` event:

1. Identify the customer, offer id, start time, offer type, and duration.
2. Define the offer expiration time as `start time + duration in hours`.
3. Search that customer's events inside the validity window.

4. Mark BOGO and discount offers successful if the customer viewed and completed the same offer.
5. Mark informational offers successful if the customer viewed the offer and then made a transaction.

Modeling

The initial candidate models will include:

- Majority-class dummy classifier.
- Logistic regression.
- Random forest classifier.

Categorical variables will be one-hot encoded. Numeric features will be scaled for logistic regression. The random forest will be tuned with grid search and cross-validation.

Validation

The final model will be selected based on validation performance and then evaluated once on the held-out test set. The analysis will include:

- Model comparison table.
- Classification report.
- Confusion matrix.
- Feature importance analysis.
- Segment-level response rates by offer type, age group, gender, and income group.

Expected Outcome

The expected outcome is a model that outperforms the majority-class benchmark on F1 score and provides interpretable insight into which customers are most likely to respond to each type of Starbucks offer.

The project will also discuss limitations, including the simulated nature of the dataset and the distinction between predicting response and estimating causal uplift.